R5. Turning Store Chatter Into Action: The Voice-of-Store Engine

Real-world pain: Head office is drowning in unstructured feedback — customer reviews, app-store comments, store-staffWhatsApp notes, complaint tickets. Real operational problems (a broken freezer in one store, a mislabelled allergen, a rude billing counter) are buried in noise and surface weeks late, if ever.

The problem: Convert a messy multi-source, multilingual feedback stream into a prioritised weekly action list, routed to the right owner.

What we provide:

- 40k feedback records from four channels with genuinely different writing styles: online reviews, in-app ratings with comments, store-staffshifl notes, and call-centre ticket transcripts
- Code-mixed text (English, Hinglish, transliterated Tamil/Marathi) with typos, emojis, and truncation
- Store master and a taxonomy of ~30 operational issue types with the owning function (Store Ops, Supply Chain, Merchandising, Digital, HR)
- Hidden labels on a sample: issue type, severity, and whether it was "actionable"

Assumptions students may make:

- Feedback is unverified; a single angry review is not proof of a systemic issue
- Volume differs hugely by store size, so raw counts must be normalised
- Anything mentioning safety, allergens, hygiene, or staffmisconduct is high-severity by policy, regardless of volume
- No audio is provided (transcripts only)

Desired output:

1. A weekly one-page brief per region: top 5 issues, evidence quotes, affected stores, trend arrow vs. last week
2. Routing: each issue tagged with owning function and a severity level, with the escalation rule stated
3. **Emerging-issue detection** — the killer feature: flag an issue that is small in absolute volume but rising sharply at one store, before it becomes a national problem
4. `evaluate.py` score = classification quality on hidden labels + detection lead-time on three seeded "planted" incidents hidden in the data (this is where teams genuinely differentiate)
5. A precision statement: of the issues you escalated, what fraction were truly actionable

Stretch: a de-duplication layer so the same incident reported through four channels appears once.